TEHRD – TRUST'S

KISHKINDA UNIVERSITY

(Established under the Karnataka State Act No.20 of 2023 | Recognized by Govt. of Karnataka & UGC, New Delhi)

USN | | | | | | | | | | | **Course Code** | 2 | 3 | C | S | 3 | 2 |

Third Semester B.Tech Degree Examinations

DIGITAL SYSTEM DESIGN and COMPUTER ORGANISATION

Duration: 3 hrs **Max. Marks: 100**

Note : **1**. Answer any **FIVE** full questions, choosing **ONE full** question from each module
2. Missing data, if any, may be suitably assumed.

Question Number		Question	Marks	(RBTL:CO:PI)
		Module-I		
1	a	List out the K-Map rules for grouping the Boolean equations and explain with an example?	5	(3 :1: 2.1.2)
	b	Determine the Prime Implicant Using Quine- McCluskey Method for the following f (a, b, c, d) = $\sum_m$ (0,1,2,5,6,7,8,9,10,14)	10	(1 :1: 1.4.1)
	c	Place the following equations in to proper canonical form: $T = f(A,B,C,D) = (A + \bar{B} + C)(\bar{A} + D)$	5	(1 :1: 1.4.1)
		OR		
2	a	Simplify the following equation using the four variable K-Map: Y= f (a, b, c, d) = $\sum_m$ (1,3,4,5,10,12,13)	10	(2 :1: 2.1.2)
	b	Using the Map Entered Variables Method, for four variable maps find the minimum sum-of-products for the following expression? f (a, b, c, d) = $\sum_m$ (2,5,6,9) + $\sum_d$ (1,3,4,13,14)	10	(3 :1: 2.1.2)
		Module-II		
3	a	Design and explain 8:1 Multiplexer with truth table and draw the circuit diagram?	6	(3 :2: 1.4.1)
	b	Design and explain 8:3 Encoder with truth table and draw circuit diagram?	6	(1 :2: 1.4.1)
	c	Design and explain 3:8 Decoder with truth table and draw circuit diagram?	8	(6 :2: 2.4.1)
		OR		
4	a	Explain JK- Flip Flop with neat diagram and Function table	6	(3 :2: 2.4.1)

Model Question Paper

Note: RBTL **(Revised Bloom's Taxonomy Level), CO (Course Outcome), PI (Performance Indicator)**

	b	Explain the Programmable Logic Array with an example. Explain the Programmable Array Logic with an example.	5 5	(3 :2: 2.4.1)
	c	Define Latch and differentiate between SR Latch and Gated SR Latch?	4	(2 :2: 1.4.1)
		Module-III		
5	a	Design a Mod- 12 Synchronous Counter using JK-Flip flop with timing wave forms?	10	(2 :3: 1.4.1)
	b	Explain the operation of SISO and PIPO Shift registers with neat wave forms?	10	(2 :3: 1.4.1)
		OR		
6	a	Define Register and Explain the Register Transfer with neat diagram?	10	(2 :3: 1.3.1)
	b	Design a Mod 8 Synchronous Binary Counter using SR-Flip flop with timing wave form?	10	(4 :3: 1.4.1)
		Module-IV		
7	a	Explain the following concepts: a). Clock Rate b). Performance measurement	10	(1 :3: 1.4.1)
	b	What is addressing mode? List out the addressing modes and explain any three with examples?	10	(2 :4: 2.1.2)
		OR		
8	a	What are the Instructions used for different types of operations and explain about each in detail?	10	(2 :4: 2.1.2)
	b	With a neat block diagram explain the single Bus Structure.	10	(2 :4: 2.1.2)
		Module-V		
9	a	Explain how DMA Controllers are used in a computer system with a neat diagram?	10	(2 :4: 2.1.2)
	b	Explain the concept of Array multiplication of positive binary operands?	10	(6 :4: 2.1.2)
		OR		
10	a	Illustrate both addition and subtraction of signed numbers with examples in 2's complement representation.	10	(2 :5: 1.4.1)
	b	Explain Booth algorithm scheme in signed operand Multiplication?	10	(6 :5: 2.1.2)

Model Question Paper

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